

Daily Work Report

Employee Name: Atharva Ravikiran Avhad

Date: 24/07/2026

Reporting Manager: Aditya Chauhan

Work Summary

Started Week 3 objectives focusing on transforming the PDF extraction service into a production-ready microservice. Successfully established a comprehensive automated testing framework to prevent regressions. Additionally, enhanced the application's stability and observability by implementing custom domain-specific exception handling, structured JSON logging, and threadpool-based asynchronous processing to handle heavy PDF extractions without blocking the main event loop

Completed Tasks

- **Automated Test Suite (Unit & API):** Wrote and executed 14 tests using pytest and FastAPI's TestClient, covering edge cases like empty uploads, invalid extensions, corrupted files, and zero-page PDFs. All tests passed successfully.
- **Structured Error Handling:** Created custom exception classes (PDFCorruptedException, PDFPasswordProtectedException) to provide cleaner, more specific error routing within the microservice.
- **Asynchronous Processing:** Integrated FastAPI's run_in_threadpool to push CPU-heavy extraction tasks into background threads, preventing HTTP request blocking.
- **Structured JSON Logging:** Implemented a custom JSON formatter for the system logger to allow seamless integration with centralized production logging systems.

Pending Tasks

- Benchmark extraction pipeline using 10 different PDF categories.
- Analyze extraction failures and classify root causes.
- Profile system performance (CPU, RAM, Execution Time).

- Optimize extraction pipeline based on profiling metrics.
- Prepare final engineering reports (Performance Profiling, Failure Analysis, Production Readiness Checklist).

Plan for Tomorrow

- Curate a dataset of 10 diverse PDFs (financial, legal, invoices, etc.) and write a benchmarking script to process them.
- Record success rates, extraction accuracy, and processing times.
- Perform failure analysis on any failed extractions and classify the root causes for the engineering report.

Employee Declaration

I hereby confirm that the above information accurately represents the work completed by me on the mentioned date.

Employee Signature:

Atharva Avhad